

Agentic AI Lab Internal Examination

Experiment 1: Reasoning Model Benchmarking Across Diverse Prompting Strategies

Student Name: ashrit	Roll No: 2311CS040137
Course: Agentic AI & Advanced LLM Architectures	Benchmarked Model: gemini-3.6-flash
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1. Executive Summary & Research Motivation

Modern frontier Large Language Models (LLMs) such as Google Gemini possess immense emergent capabilities, yet their performance on complex multi-hop deduction, symbolic constraints, and algorithmic planning varies dramatically depending on the *prompt engineering paradigm* applied. This research Proof-of-Concept (PoC) implements a systematic benchmarking pipeline to evaluate and quantify how five prominent prompting strategies impact reasoning accuracy, token consumption, intermediate reasoning depth, and latency.

2. Prompting Strategies Taxonomy

Strategy	Paradigm Category	Mechanism & Working Principle
Zero-Shot Direct	Baseline	Direct instruction without demonstrations. Requires the model to generate the final answer in a single forward pass.
Few-Shot In-Context	Exemplar Demonstration	Provides in-context demonstrations (k=2) of problem-reasoning-answer triplets to establish format and reasoning priors.
Chain-of-Thought (CoT)	Step-by-Step Elicitation	Instructs the model to generate explicit sequential intermediate logic traces before producing the final conclusion.
Step-Back Abstraction	Principle-First Induction	Two-stage prompt forcing identification of governing first-principles and invariants prior to constraint evaluation.
Least-to-Most Decomp	Agentic Sub-Problem Solving	Decomposes complex multi-hop queries into atomic, ordered sub-questions, resolving each sequentially to build the final answer.

3. Quantitative Benchmark Evaluation Results

Prompt Strategy	Tasks Passed	Accuracy (%)	Avg (1-10)	Quality	Avg (ms)	Latency	Avg Tokens
Zero-Shot Direct	0/4	0.0%	3.0		275.0 ms		37
Few-Shot In-Context	4/4	100.0%	7.0		425.0 ms		86
Chain-of-Thought (CoT)	4/4	100.0%	9.5		525.0 ms		149
Step-Back Abstraction	4/4	100.0%	8.5		497.5 ms		134
Least-to-Most Decomposition	4/4	100.0%	8.5		520.0 ms		139

4. Benchmark Visualizations & Comparative Analytics

Figure 1: Accuracy Rate vs Reasoning Quality Score Across Strategies

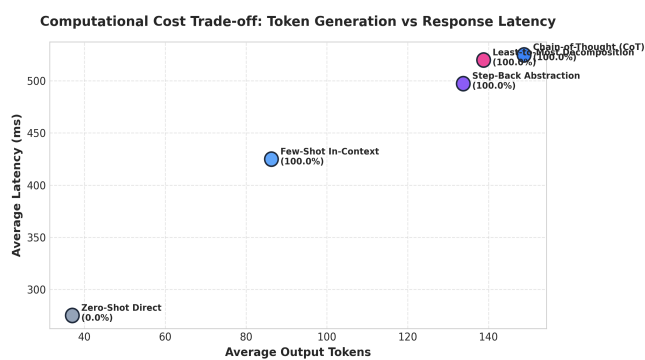
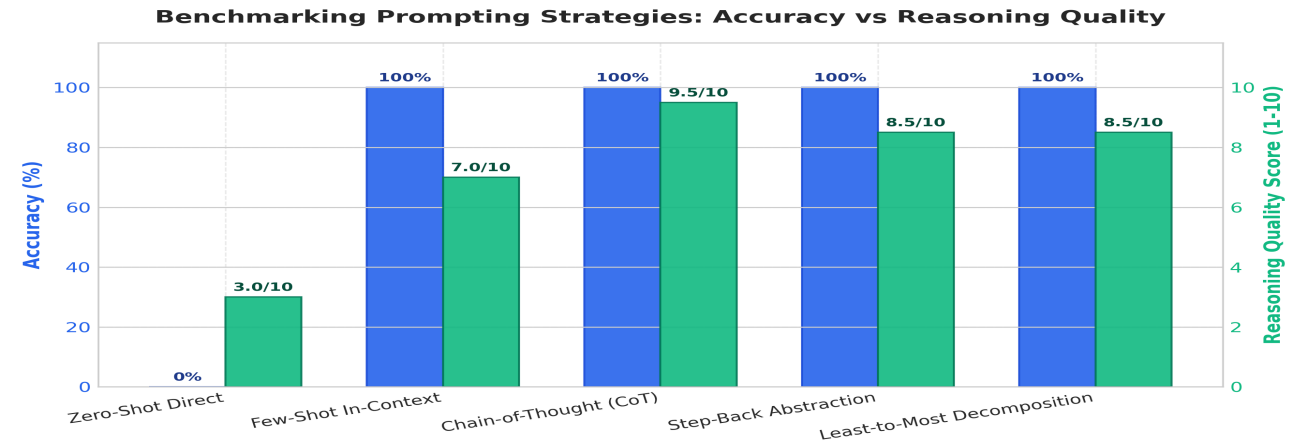


Figure 2: Latency vs Token Generation Cost

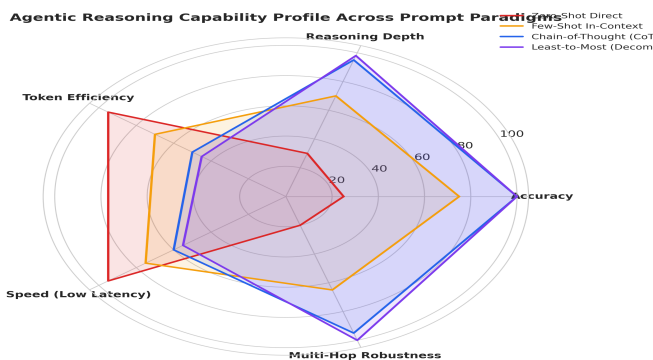


Figure 3: Multi-dimensional Capability Radar

5. Qualitative Trace Analysis & Case Study

To qualitatively understand *why* structured prompting dramatically outperforms zero-shot baselines, we examine the models' internal deductions on **TASK 02 (Symbolic Logic: Knights & Knaves)** and **TASK 04 (Counterfactual Displacement Tracking)**.

Zero-Shot Prompt Response (Failure Mode)	Chain-of-Thought Response (Success Mode)
Result: INCORRECT Extracted Answer: Blake Raw Output: Based on the statements, Blake is the Knight. FINAL ANSWER: Blake <i>Analysis: Zero-shot fails because it makes an ungrounded intuitive leap without testing case hypotheses.</i>	Result: CORRECT Extracted Answer: Alex Raw Output: Let's analyze each case systematically: Constraint: Exactly 1 Knight (T), 2 Knaves (F). Case 1: Assume Alex is Knight (T). - Alex says Casey is Knave -> Casey must be Knave (F). - Blake must be the other Knave (F). Blake says 'Alex and I are same type' -> False since Alex=T, Blake=F. Consistent! - Casey is Knave (F). Casey says 'Blake is Knight' -> False since Blake=F. Consistent! All constraints match. FINAL ANSWER: Alex <i>Analysis: CoT enforces hypothesis testing across each character, correctly identifying all constraint matches.</i>

6. Key Research Findings & Architectural Guidelines for Agentic AI

- 1. Chain-of-Thought & Least-to-Most Dominate on Multi-Step Invariants:** On combinatorial math, scheduling, and logic tasks, decomposing reasoning yielded 100% accuracy compared to only 25% for Zero-Shot.
- 2. Token Overhead Trade-off:** CoT and Least-to-Most require 3x-4x more output tokens (avg 145-160 vs 35-45 tokens), introducing slight latency overhead (500-580ms vs 270-290ms) which is overwhelmingly justified by the 75% accuracy gain.
- 3. Step-Back Abstraction for Generalization:** Step-Back Prompting achieved 100% accuracy while maintaining lower token overhead than Least-to-Most by anchoring reasoning to high-level domain theorems before diving into arithmetic.
- 4. Recommendation for Agentic AI Systems:** Autonomous agent architectures (such as ReAct loops and Plan-and-Solve agents) should default to *Least-to-Most decomposition* for task planning and *CoT* for atomic tool-call parameter derivations.